BGIN Block #9 Meeting Report

- Rebuilding Value, Governance and Reputation on Blockchain -

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The Global Network for Blockchain Stakeholders™

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3.10. DTWG Interactive Workshop Session 4: Smart Contract Security, Governance, and Impact of AI

Date: Nov 20, 2023
Time: 15:30 - 17:00
Session Chair: Ellie Rennie (RMIT)

Paper Summary

- The major change happening is massive numbers of developers started using copilot provided by microsoft which are used to code smart contracts. AI has been seen as another tool to write codes. We could tell from the increasing number of codes written by AI in github. Emergence of AI is helping not just engineers to code, but people of non-technical background to read and understand what smart contracts are doing inside the codes.

Discussion points

- Smart contract engineers were very thin on the ground several years ago, but codified taxonomies have arisen as a part of their emerging professionalism. It has got a baseline, a useful set of inputs for AI, in a programmatic sense.

- Can we use blockchain technology to help with the governance of AI models? Is there a way for these AI models to upload on-chain attestation? If one day OpenAI gives you a model that is widely open source, billions of numbers of data will be in a CSV file, which is hard to read even if you have access.

- In terms of using AI to read smart contract data, smart contract data today in EVM environments get compiled down into byte code, which is interpreted by decompiler and it is abstracted in the way that only certain engineers with the knowledge can understand. We can eventually extract common elements into non technical and everyone can be a programmer of smart contracts.

- Running simulations is a large part of building proper smart contract systems. In the context of layer one blockchain network, they have deep testing teams that spin up 100+ node networks, running different transactions, dispatching patterns over those networks, mutating the infrastructure, etc. There’s quite a lot of scope for having an AI augmented simulation.

- Is AI going to be a standardization execution tool or a standard setting tool? If it is a standard setting tool, are we going to create a risk where everyone is using old traditional inputs that don't have creativity? Should there be a focus on AI using the standards that humans have created as an execution? If yes, then we should

avoid having automated smart contract creation agents since automated smart contract agents are actually limiting our ability.

- In the blockchain and AI spaces, trying to prescribe how things should be done has not really worked out so far. If we want to make it happen, there should be a coalition of blockchain foundations working together, combining resources, and making it happen from within the industry. If people in the blockchain space are going to use AI, they need to use it in the right way. Blockchain foundation should be proactive and take ethics more seriously generally.